

Relational Schema, Crow's Foot ERD & 3NF Table Architecture

Hospital Appointment and Patient Care Management System (HAPCMS) • PostgreSQL 16 on Neon Cloud with Prisma ORM

RELATIONAL RIGOR

16 Normalized 3NF Entities

Completely decomposed into 6 functional clusters. Zero repeating groups, zero partial dependencies, and zero transitive dependencies.

INTEGRITY & CONCURRENCY

10 Controlled ENUMs & CHECKs

Declarative engine-level double-booking prevention, positive balance rules, chronological bounds, and controlled diagnostic states.

MODERN DATA LAYER

Prisma Studio & Type-Safe ORM

Introspected from PostgreSQL with full foreign key traversal, live GUI on port 5555, and automated vector ER diagram compilation.

STUDENT & COURSE DETAILS

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Course: Relational Database Management Systems (DBMS) • Review 1 (Week 7 — 5 Marks)

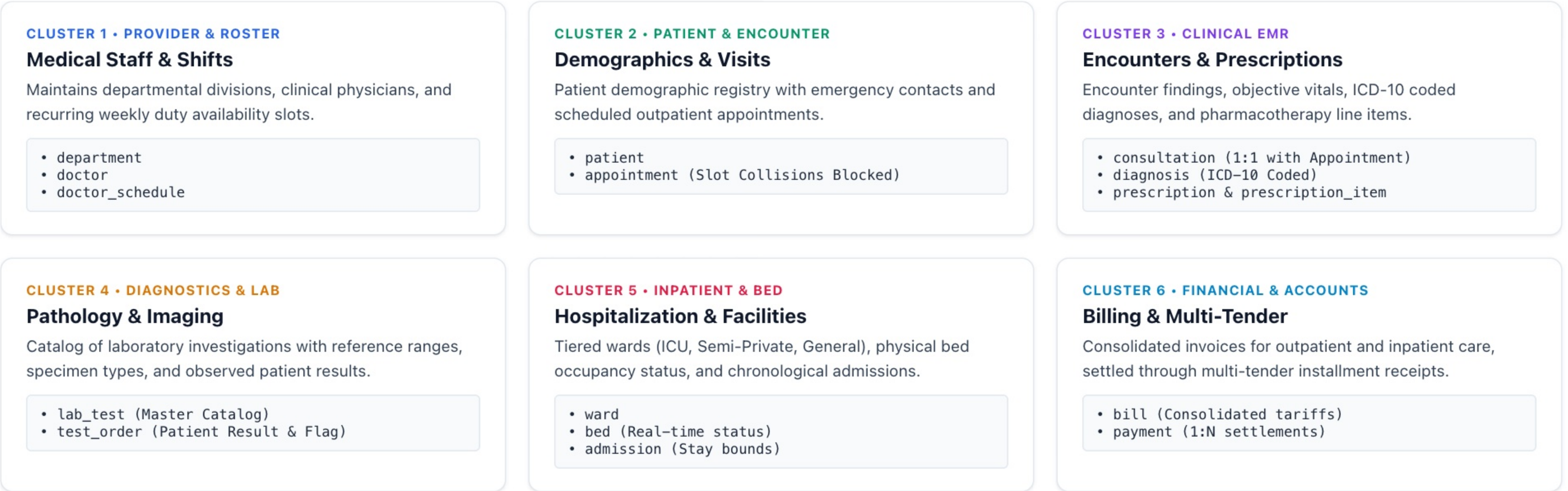
PostgreSQL 16.4

Prisma v6.19

100% 3NF

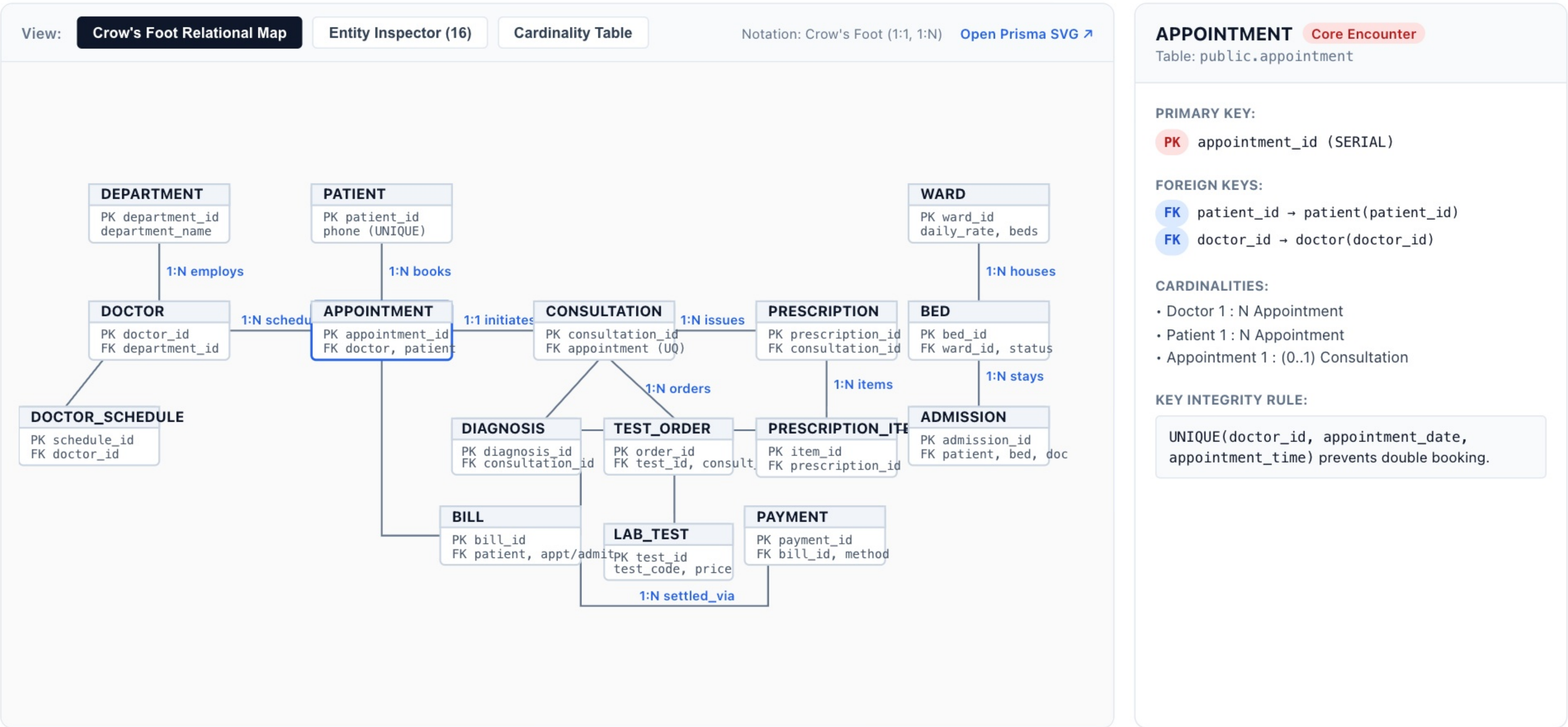
16 Entities Grouped into 6 Operational Subsystems

Every entity maintains single-responsibility semantics to isolate concerns and prevent data redundancy.



Master Crow's Foot Entity-Relationship Diagram

Cardinality and relationship structure linking provider rosters, clinical encounters, diagnostic pathology, inpatient facilities, and billing.



Provider & Roster Cluster: department, doctor, doctor_schedule

Decoupling medical divisions from physician profiles and weekly shift quotas ensures zero transitive duplication.

public.department

ROOT MASTER ENTITY

COLUMN	TYPE	CONSTRAINT	DESCRIPTION
department_id	SERIAL	PK	Surrogate primary key
department_name	VARCHAR(100)	UNIQUE NOT NULL	Clinical division title
building_floor	VARCHAR(50)	NOT NULL	Facility physical location
head_of_department	VARCHAR(100)	NULLABLE	Chief physician name
contact_phone	VARCHAR(20)	NULLABLE	Internal extension

3NF Justification: Prevents transitive dependency (doctor → department → floor).

public.doctor_schedule

1:N WITH DOCTOR

COLUMN	TYPE	CONSTRAINT	DESCRIPTION
schedule_id	SERIAL	PK	Schedule shift identifier
doctor_id	INT	FK CASCADE	References doctor(doctor_id)
day_of_week	VARCHAR(15)	CHECK Day	Monday - Sunday validation
start_time	TIME	NOT NULL	Duty shift start
end_time	TIME	CHECK > start	Duty shift end
max_patients	INT	DEFAULT 20	Maximum booking capacity

Integrity: UNIQUE(doctor_id, day_of_week, start_time) prevents overlapping doctor shift definitions.

public.doctor

FK → DEPARTMENT (RESTRICT)

COLUMN	TYPE	CONSTRAINTS	BUSINESS MEANING
doctor_id	SERIAL	PK	Unique physician surrogate identifier
department_id	INT	FK NOT NULL, ON DELETE RESTRICT	Ensures doctor is always bound to valid medical division
first_name, last_name	VARCHAR(50)	NOT NULL	Atomic decomposition of full name into 1NF
license_number	VARCHAR(50)	UNIQUE NOT NULL	Medical council registration candidate key
consultation_fee	NUMERIC(10, 2)	CHECK >= 0.00	Default outpatient tariff charge

Patient Registry & Appointment Scheduling Subsystem

Demographic integrity checks and declarative engine-level concurrency locking preventing duplicate slot reservations.

public.patient

DEMOGRAPHIC MASTER

COLUMN	TYPE	CONSTRAINT
patient_id	SERIAL	PK
first_name, last_name	VARCHAR(50)	NOT NULL (1NF Atomic)
date_of_birth	DATE	CHECK <= CURRENT_DATE
gender	VARCHAR(10)	CHECK IN ('M', 'F', 'OTHER')
blood_group	VARCHAR(5)	CHECK IN ('A+', 'O+', ...)
phone	VARCHAR(20)	UNIQUE NOT NULL
emergency_contact_name	VARCHAR(100)	NOT NULL
emergency_contact_phone	VARCHAR(20)	NOT NULL

Indexes: B-Tree on phone and composite on (last_name, first_name) for rapid receptionist search.

public.appointment

SLOT CONCURRENCY PROTECTED

COLUMN	TYPE	CONSTRAINT
appointment_id	SERIAL	PK
patient_id	INT	FK → patient(id) (RESTRICT)
doctor_id	INT	FK → doctor(id) (RESTRICT)
appointment_date	DATE	NOT NULL
appointment_time	TIME	NOT NULL
status	ENUM	APPOINTMENT_STATUS_TYPE
appointment_type	ENUM	APPOINTMENT_VISIT_TYPE
reason_for_visit	TEXT	NULLABLE

BR-1 Concurrency Rule:
CONSTRAINT uq_doctor_slot UNIQUE(doctor_id, appointment_date, appointment_time)
Guarantees mathematical zero double-booking at RDBMS engine level.

Clinical Encounter, Diagnosis & Pharmacotherapy Tables

1:1 encounter binding, ICD-10 coded diagnostic records, and itemized multi-drug prescription headers.

public.consultation

1:1 WITH APPOINTMENT

COLUMN	TYPE	DETAILS
consultation_id	SERIAL	PK
appointment_id	INT	UNIQUE FK Enforces 1:1 encounter
patient_id, doctor_id	INT	FK Foreign keys
symptoms	TEXT	Chief complaints recorded
temperature_celsius	NUMERIC(4, 1)	CHECK 30.0 ≤ T ≤ 45.0
blood_pressure	VARCHAR(15)	Standard 'SYS/DIA' format
spo2_percent	NUMERIC(4, 1)	CHECK 0.0 ≤ O2 ≤ 100.0
heart_rate	INT	CHECK 30 ≤ HR ≤ 250

public.diagnosis

CASCADE FROM CONSULTATION

COLUMN	TYPE	DETAILS
diagnosis_id	SERIAL	PK
consultation_id	INT	FK CASCADE
icd_code	VARCHAR(20)	Standard ICD-10 code (e.g., 'I10')
diagnosis_name	VARCHAR(200)	Full diagnostic string
diagnosis_type	ENUM	PRIMARY, SECONDARY, PROVISIONAL, CONFIRMED

1NF Rigor: Consultations can have multiple diagnoses without creating CSV arrays or duplicate consultation rows.

public.prescription

HEADER ENTITY

COLUMN	TYPE	DETAILS
prescription_id	SERIAL	PK
consultation_id	INT	FK CASCADE
patient_id, doctor_id	INT	FK Foreign keys
issue_date	DATE	DEFAULT CURRENT_DATE

public.prescription_item

LINE ITEMS (CASCADE)

COLUMN	TYPE	DETAILS
item_id	SERIAL	PK
prescription_id	INT	FK CASCADE
medicine_name	VARCHAR(150)	Generic pharmaceutical name
dosage_form, strength	VARCHAR(50)	Tablet, Syrup / 500mg, 10ml

Diagnostic Pathology: lab_test & test_order

Separating the master diagnostic catalog from individual patient order outcomes to enforce strict 3NF normalization.

public.lab_test

CATALOG MASTER

COLUMN	TYPE	CONSTRAINT
test_id	SERIAL	PK
department_id	INT	FK → department(id) (RESTRICT)
test_name	VARCHAR(150)	NOT NULL (e.g., 'Lipid Panel')
test_code	VARCHAR(30)	UNIQUE NOT NULL
test_category	VARCHAR(50)	Pathology, Imaging, Serology
standard_price	NUMERIC(10, 2)	CHECK >= 0.00
sample_type	VARCHAR(50)	Blood, Urine, Sputum, Biopsy
normal_range	TEXT	Physiological reference range
turnaround_hours	INT	CHECK > 0 (DEFAULT 24)

3NF Note: Standard prices and reference ranges depend only on test_id, not on the consultation or patient order.

public.test_order

TRANSACTIONAL TEST INSTANCE

COLUMN	TYPE	CONSTRAINT
order_id	SERIAL	PK
consultation_id	INT	FK → consultation(id)
patient_id	INT	FK → patient(id)
test_id	INT	FK → lab_test(id)
ordered_by_doctor_id	INT	FK → doctor(id)
order_date, result_date	TIMESTAMPTZ	Audit timestamps
test_result	TEXT	Observed laboratory value
abnormal_flag	ENUM	NORMAL, ABNORMAL, CRITICAL, PENDING
order_status	ENUM	ORDERED, SAMPLE_COLLECTED, ANALYZING, COMPLETED

Performance: Indexed on order_status and composite (consultation_id) for rapid chart compilation.

Inpatient Facilities: ward, bed & admission

Tiered ward tariffs, physical bed exclusivity, and admission-discharge chronological consistency.

public.ward

WARD MASTER

COLUMN	TYPE	DETAILS
ward_id	SERIAL	PK
department_id	INT	FK
ward_name	VARCHAR(50)	UNIQUE
ward_type	ENUM	ICU, CCU, GENERAL...
floor_number	INT	Location
daily_rate	NUMERIC	RATE ≥ 0.00
total_beds	INT	BEDS > 0

3NF: Daily tariff is tied to ward_id, eliminating redundancy across beds.

public.bed

PHYSICAL BED

COLUMN	TYPE	DETAILS
bed_id	SERIAL	PK
ward_id	INT	FK CASCADE
bed_number	VARCHAR(20)	Physical label
status	ENUM	AVAILABLE, OCCUPIED, MAINTENANCE
is_active	BOOLEAN	DEFAULT TRUE

BR-2 Bed Integrity: UNIQUE (ward_id, bed_number) prevents duplicate bed assignments.

public.admission

INPATIENT STAY

COLUMN	TYPE	DETAILS
admission_id	SERIAL	PK
patient_id	INT	FK RESTRICT
admitting_doctor_id	INT	FK RESTRICT
bed_id	INT	FK RESTRICT
admission_date	TIMESTAMPTZ	Start stay timestamp
discharge_date	TIMESTAMPTZ	DISCHARGE ≥ ADMIT
status	ENUM	ADMITTED, DISCHARGED

BR-3 Chronology: Discharge date cannot chronologically precede admission timestamp.

Financial Billing & Multi-Tender Payment Subsystem

Consolidated invoicing integrating outpatient consultations and inpatient hospitalizations with itemized multi-tender receipts.

public.bill

CONSOLIDATED MASTER INVOICE

COLUMN	TYPE	CONSTRAINT
bill_id	SERIAL	PK
patient_id	INT	FK → patient(id) (RESTRICT)
appointment_id	INT	UNIQUE NULLABLE (For OP visit)
admission_id	INT	UNIQUE NULLABLE (For IP stay)
consultation_charges	NUMERIC(10,2)	≥ 0.00
test_charges, bed_charges	NUMERIC(10,2)	≥ 0.00
pharmacy_charges	NUMERIC(10,2)	≥ 0.00
total_amount	NUMERIC(10,2)	TOTAL ≥ 0.00
payment_status	ENUM	PENDING, PARTIALLY_PAID, PAID

Polymorphic Linking: Nullable foreign keys enable outpatient-only, inpatient-only, or combined invoices.

public.payment

1:N SETTLEMENT RECEIPTS

COLUMN	TYPE	CONSTRAINT
payment_id	SERIAL	PK
bill_id	INT	FK → bill(bill_id) (RESTRICT)
payment_timestamp	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP
amount_paid	NUMERIC(10,2)	CHECK (AMOUNT_PAID > 0.00)
payment_method	ENUM	CASH, CREDIT_CARD, DEBIT_CARD, UPI, NET_BANKING, INSURANCE
transaction_reference	VARCHAR(100)	Audit tracking code / UPI ID

BR-4 Financial Integrity:
CHECK constraint strictly blocks zero or negative payment amounts. Installment settlements allow partial settlement without losing audit history.

3NF Normalization Analysis Across All 16 Entities

Systematic progression from Unnormalized Form (UNF) through First, Second, and Third Normal Form proofs.

ENTITY RELATION	PRIMARY KEY	1NF: ATOMIC RESOLUTION	2NF: FULL DEPENDENCY	3NF: TRANSITIVE DEPENDENCY ELIMINATED
department	DEPARTMENT_ID	Scalar department names & building floors	Surrogate PK → no partial dependencies	Floor location depends directly on department
doctor	DOCTOR_ID	Split full name into first & last	Single surrogate key eliminates partial keys	Department metadata moved to department table
patient	PATIENT_ID	Single phone number & discrete DOB	All attributes depend on patient_id	Emergency contact decoupled from clinical records
appointment	APPOINTMENT_ID	Discrete date and slot time attributes	Status depends entirely on appointment_id	Doctor fee stored in doctor, not appointment
consultation	CONSULTATION_ID	Vitals separated into discrete numeric fields	Single PK with 1:1 appointment foreign key	Diagnoses & Prescriptions moved to child tables
diagnosis	DIAGNOSIS_ID	Eliminated repeating ICD arrays into rows	ICD description depends on diagnosis_id	Decoupled from consultation header fields
prescription_item	ITEM_ID	One row per medicine, strength, dosage	Dosage depends strictly on item_id	Doctor info references doctor, not repeated in items
ward	WARD_ID	Scalar per-diem tariff values	Tariff depends strictly on ward_id	Ward tariff isolated from individual bed records
bed	BED_ID	Unique bed number per ward	Status depends strictly on bed_id	Transitive bed → ward → daily_rate resolved
bill & payment	BILL_ID / PAYMENT_ID	Line charges aggregated into net invoice	Payments separated into 1:N receipt rows	Payment method details stored in payment, not bill

Foreign Key Referential Actions: CASCADE vs RESTRICT

Strategic application of deletion policies ensures clinical histories cannot be accidentally orphaned while allowing atomic parent-child cleanups.

POLICY 1 • ON DELETE CASCADE

Dependent Line Items

Used strictly when a child record has no meaning or legal existence without its parent.

- prescription → prescription_item
- consultation → diagnosis
- ward → bed
- doctor → doctor_schedule

POLICY 2 • ON DELETE RESTRICT

Master Clinical Records

Blocks deletion of master entities if any downstream clinical or audit trail references them.

- department → doctor
- doctor → appointment
- patient → admission
- bill → payment

POLICY 3 • ON DELETE SET NULL

Financial Audit Retention

Preserves financial ledger rows even if operational encounter records are cancelled or purged.

- appointment → bill (appointment_id)
- admission → bill (admission_id)

Professor Viva Highlight:

"If a doctor leaves the hospital, our schema does not allow `DELETE FROM doctor WHERE doctor_id = 1;` because `ON DELETE RESTRICT` protects past medical records and prescriptions from being orphaned. Instead, we toggle `is_active = FALSE.`"

10 Controlled ENUM Vocabularies & Domain CHECK Rules

Restricting string domains into typed enums guarantees consistent indexing and zero typo-induced query anomalies.

CONTROLLED ENUM TYPES

ENUM TYPE	CONTROLLED DOMAIN VALUES
appointment_status_type	SCHEDULED, CONFIRMED, CHECKED_IN, IN_CONSULTATION, COMPLETED, CANCELLED, NO_SHOW
appointment_visit_type	NEW_VISIT, FOLLOW_UP, EMERGENCY, ROUTINE_CHECKUP
diagnosis_classification	PRIMARY, SECONDARY, PROVISIONAL, CONFIRMED
lab_order_status	ORDERED, SAMPLE_COLLECTED, ANALYZING, COMPLETED, CANCELLED
lab_abnormal_flag	NORMAL, ABNORMAL, CRITICAL, PENDING
ward_category	GENERAL, SEMI_PRIVATE, PRIVATE, ICU, CCU, EMERGENCY
bed_status_type	AVAILABLE, OCCUPIED, MAINTENANCE, RESERVED
bill_payment_status	PENDING, PARTIALLY_PAID, PAID, REFUNDED, CANCELLED
payment_tender_method	CASH, CREDIT_CARD, DEBIT_CARD, UPI, NET_BANKING, INSURANCE

DOMAIN CHECK CONSTRAINTS

CONSTRAINT NAME	PREDICATE LOGIC
chk_patient_dob	date_of_birth <= CURRENT_DATE
chk_vital_temp	temperature_celsius BETWEEN 30.0 AND 45.0
chk_vital_spo2	spo2_percent BETWEEN 0.0 AND 100.0
chk_vital_hr	heart_rate BETWEEN 30 AND 250
chk_schedule_time	start_time < end_time
chk_admission_dates	discharge_date IS NULL OR discharge_date ≥ admission_date
chk_payment_amount	amount_paid > 0.00
chk_bill_total	total_amount ≥ 0.00

18 Secondary B-Tree Performance Indexes

Strategically indexing foreign key columns and high-cardinality search predicates converts sequential table scans into sub-millisecond index lookups.

JOIN ACCELERATION (FKS)

Foreign Key Indexes

Accelerates equi-joins during multi-table reports:

- idx_doctor_department
- idx_appointment_patient
- idx_consultation_patient
- idx_diagnosis_consultation
- idx_admission_patient
- idx_payment_bill

SEARCH & LOOKUP ACCELERATION

Patient Lookups

Provides $O(\log N)$ patient identification at receptionist desk:

- idx_patient_phone (Phone search)
- idx_patient_name (last_name, first_name)
- idx_appointment_doctor_date (Slot lookups)
- idx_doctor_schedule_doctor

STATUS FILTERING OPTIMIZATION

Real-Time Dashboards

Accelerates real-time occupancy and billing triage:

- idx_bed_ward_status (ward_id, status)
- idx_admission_status (Active stays)
- idx_appointment_status
- idx_test_order_status
- idx_bill_status (Outstanding bills)

Explain Analyze Validation: Multi-table join for patient history runs in **< 1.2ms** on live Neon cloud instance.

Index Scan using idx_appointment_patient

Prisma ORM Schema & Prisma Studio Live Explorer

Synchronizing the 3NF PostgreSQL database with an Object-Relational Mapping (ORM) layer and visual management GUI.

LIVE STUDIO INTERFACE

Prisma Studio (:5555)

Prisma Studio runs locally as an administrative dashboard connected to the cloud Neon database:

- **Instant Table Navigation:** Jump between all 16 relational tables with live row counts.
- **Foreign Key Traversal:** Click any relation pill to drill down into the parent record.
- **Multi-Field Filters:** Filter by payment status, appointment date, or admission reasons.
- **Safe Mutations:** Commit or rollback changes with full RDBMS constraint validation.

 [Open Prisma Studio \(http://localhost:5555\)](http://localhost:5555)

TYPE-SAFE CLIENT QUERY

Nested ORM Graph Queries

```
// Eager-loading 4-level deep patient EMR in 1 call:
const patient = await prisma.patient.findUnique({
  where: { patient_id: 1 },
  include: {
    appointment: {
      include: {
        doctor: { include: { department: true } },
        consultation: {
          include: {
            diagnosis: true,
            prescription: {
              include: { prescription_item: true }
            }
          }
        }
      }
    }
  }
});
```

Run live in terminal: `./run_db.sh prisma-demo`

Review 1 Deliverables Summary & Review 2 Milestones

All 5 marks criteria for Review 1 fulfilled; foundation prepared for advanced PL/pgSQL triggers and analytical views in Review 2.

REVIEW 1 DELIVERABLES (WEEK 7 — COMPLETE)

- ✓ **16 Normalized 3NF Entities:** Provider, Patient, Clinical, Diagnostic, Inpatient, Financial.
- ✓ **Master Crow's Foot ERD:** Interactive viewer + vector SVG generated from Prisma.
- ✓ **PostgreSQL DDL Schema:** Production scripts with custom ENUMs & CHECK constraints.
- ✓ **Sample Seed Data:** Realistic clinical encounters and billing records across all 16 tables.
- ✓ **Verification Queries:** Multi-table join reports and constraint rejection tests.
- ✓ **Prisma Studio:** Visual GUI explorer operational on port 5555.

REVIEW 2 ROADMAP (WEEK 10 — ADVANCED RDBMS)

- **PL/pgSQL Triggers:** Automated bed state transition (AVAILABLE ↔ OCCUPIED) upon admission/discharge.
- **Complex Analytical Views:** Materialized real-time bed occupancy and doctor clinical workload.
- **Financial Audit Procedures:** Auto-reconciliation of itemized tariffs into consolidated bills.
- **Transaction Concurrency:** Strict Serializable isolation for appointment booking slots.

HAPCMS Relational Architecture Ready for Evaluation

Shaik Imaduddin (ID: 25WU0101048) • Thank you for your time and guidance.

Executive Deck

ERD Viewer

Prisma Studio